

Alexander (Zander) Kelley

CONTACT INFORMATION

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RESEARCH INTERESTS

computational complexity, pseudorandomness, communication complexity,
extremal and additive combinatorics, number theory

EDUCATION

University of Illinois at Urbana-Champaign

Ph.D., Computer Science, August 2024

- Dissertation: Explicit Pseudorandom Distributions for Restricted Models of Computation
- David J. Kuck Outstanding PhD Thesis Award
- Advisor: Michael A. Forbes

Texas A&M University

B.S. Computer Science, May 2017

- Honors Distinction

B.S. Mathematics, May 2017

- Honors Distinction

POSITIONS

Postdoctoral Member, Sept 2024 - Present

School of Mathematics

Institute for Advanced Study, Princeton

HONORS AND AWARDS

Frontiers of Science Award

For achievements from the past ten years that are of outstanding scholarly value.

International Congress of Basic Sciences, Beijing, China, 2025

ACM Doctoral Dissertation Honorable Mention Award

Association for Computing Machinery (ACM), 2025

David J. Kuck Outstanding PhD Thesis Award

For an outstanding doctoral thesis by a graduate student in the Department of Computer Science.

University of Illinois Urbana-Champaign, 2025

Ross J. Martin Memorial Award

For outstanding research achievement by a graduate student in the College of Engineering.

University of Illinois Urbana-Champaign, 2024

2023 Biggest Breakthroughs in Math

<https://www.youtube.com/watch?v=4HHUGnHcDQw>

Quanta Magazine, 2023

FOCS Best Paper Award

IEEE Symposium on Foundations of Computer Science (FOCS) 2023

PUBLICATIONS

Zander Kelley, Xin Lyu

More Efficient Sifting for Grid Norms, and Applications to Multiparty Communication Complexity

IEEE Symposium on Foundations of Computer Science (FOCS) 2025

Amir Abboud, Nick Fischer, Zander Kelley, Shachar Lovett, Raghu Meka

New Graph Decompositions and Combinatorial Boolean Matrix Multiplication Algorithms

ACM Symposium on Theory of Computing (STOC) 2024

Zander Kelley, Shachar Lovett, Raghu Meka
Explicit Separations Between Randomized and Deterministic Number-on-Forehead Communication
ACM Symposium on Theory of Computing (STOC) 2024

Zander Kelley, Raghu Meka
Strong Bounds for 3-Progressions (**Best Paper Award**)
IEEE Symposium on Foundations of Computer Science (FOCS) 2023

Zander Kelley, Raghu Meka
Random Restrictions and PRGs for PTFs in Gaussian Space
Computational Complexity Conference (CCC) 2022

Zander Kelley
An Improved Derandomization of the Switching Lemma
ACM Symposium on Theory of Computing (STOC) 2021

Michael A. Forbes, Zander Kelley
Pseudorandom Generators for Read-once Branching Programs, in Any Order
IEEE Symposium on Foundations of Computer Science (FOCS) 2018

Estimating the Number of Roots of Trinomials Over Finite Fields
Zander Kelley, Sean W. Owen
Journal of Symbolic Computation 2017

Zander Kelley
Roots of Sparse Polynomials Over a Finite Field
LMS Journal of Computation and Mathematics 2016

PROFESSIONAL
ACTIVITIES

Conference Reviews

STOC 2025, ITCS 2024, SODA 2024, STOC 2023, ICALP 2023, CCC 2022, STOC 2022,
STOC 2020, ICLR 2020, FOCS 2019

Journal Reviews

Theory of Computing

Invited Talks

Simons Symposium on New Frontiers in Combinatorics & Computer Science (Krün, Germany, 2025)
International Congress of Basic Science (Beijing, 2025)
Richard P. Stanley Seminar in Combinatorics (MIT, 2025)
Princeton Theory Lunch (Princeton, 2024)
Computer Science and Discrete Mathematics Seminar (Institute for Advanced Study, 2024)
Breakthroughs in Quantitative Ramsey Theory (University of Bristol, 2024)
Number Theory Seminar (University of Oxford, 2024)
Atlanta Discrete Analysis Seminar (Georgia Institute of Technology, 2023)
Shandong University Number Theory and Combinatorics Seminar (Shandong University, 2023)
Cornell Junior Theorists' Workshop (Cornell University, 2023)
Beyond the Boolean Cube (Simons Institute for the Theory of Computing, 2023)
Computer Science and Discrete Mathematics Seminar (Institute for Advanced Study, 2023)

TEACHING
EXPERIENCE

Graduate Teaching Assistant

University of Illinois Urbana-Champaign

Fall 2018 - Fall 2023

- CS473: Algorithms
- CS374: Introduction to Algorithms
- CS579: Computational Complexity

Undergraduate Teaching Assistant

Texas A&M University

Fall 2015

- MATH170: Calculus Retention Lab